UniTexEditor - Usage Guide

A lightweight texture editing tool for Unity Editor, supporting color correction, compositing, and image processing.

Features

- **Real-time preview**: Instantly visualize parameter changes
- **Multi-language support**: Switch between Japanese and English
- **GPU-accelerated**: Fast image processing via Compute Shaders
- **Non-destructive editing**: Original image is preserved throughout editing

Installation

1. Place this folder under `Assets/Editor/` in your Unity project
2. Open the editor from the Unity menu: `dennokoworks > UniTex Editor`

Basic Workflow

1. Open the editor

Select `dennokoworks > UniTex Editor` from the menu bar to open the editor window.

2. Load a texture

In the **Input** section, drag and drop the texture you want to edit into the "Source Texture" field.

3. Edit

Enable each section and adjust its parameters. Results are displayed in real time in the preview area.

4. Save

Click the **Apply & Save** button at the bottom of the window to save the result.

Feature Reference

Input

- **Source Texture**: The image to edit
- **Global Mask**: A mask to restrict the processing area (white = apply, black = skip)
  - Invert Mask: Swap the white and black of the mask
  - Mask Strength: Controls how strongly the mask is applied
  - Save Inverted: Save the inverted mask as a separate file

Color Correction

Adjusts the overall color of the image.

- **Hue**: Rotate the hue (-180° to 180°)
- **Saturation**: Color vividness (0 = grayscale, 1 = original, 2 = double)
- **Brightness**: Lightness (0 = black, 1 = original, 2 = double)
- **Gamma**: Mid-tone brightness correction (0.1 to 3.0)
- **Color Blend**: Blend a solid color into the image
    - Target Color: The color to blend
    - Blend Mode: Compositing method (Normal, Multiply, Add, Screen, Overlay)
    - Opacity: Blend strength

Tone Curve

Fine-tune brightness and individual channels using curves.

- **RGB**: Curve applied to all channels uniformly
- **Red / Green / Blue**: Per-channel curves

Blend

Composite another texture on top of the source.

- **Blend Texture**: The image to layer on top
- **Blend Mask**: A mask to restrict the compositing area
- **Blend Mode**: Compositing method
- **Opacity**: Blend strength
- **Tiling**: Repeat the texture
- **Scale**: Scale the texture
- **Offset**: Shift the texture position

Levels

Adjust input/output levels to control contrast.

- **Input Levels**: Set the black point and white point of the image
- **Midtones (Gamma)**: Adjust the brightness of mid-range values
- **Output Levels**: Clamp the output range

Sharpen / Blur

Apply edge enhancement or blur effects.

- **Mode**: Choose between Sharpen (edge enhancement) and Blur (softening).

Sharpen Mode

Uses Unsharp Mask to enhance edges.

Parameter	Range	Description
Strength	0 ~ 2	Detail amplification per pass. 1.0 is standard sharpening; 2.0 is aggressive.
Kernel Size	3 / 5 / 7 / 9	Size of the reference blur. Larger values detect edges over a wider area.
Iterations	1 ~ 8	Number of times the pass is repeated. Each iteration further sharpens edges. 1–2 is usually sufficient.

> **Note**: High strength or many iterations can produce halo artifacts (bright/dark fringes around edges).

Blur Mode

Applies a Gaussian blur to soften the image.

Parameter	Range	Description
Strength	0 ~ 5	Gaussian sigma — directly controls the spread of the blur per pass. Higher values produce stronger blur.
Kernel Size	3 / 5 / 7 / 9	Sampling range. A larger kernel relative to the sigma improves quality.
Iterations	1 ~ 8	Number of passes. Repeating n times is equivalent to multiplying sigma by √n.

> **Quick reference**: Light blur → Strength 0.5–1.0 / Iterations 1. Strong blur → Strength 2–3 / Iterations 3–5.

Channel Mixer

Remap the contents of RGBA channels.

- **Red / Green / Blue / Alpha Output**: Select the source channel for each output channel

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Preset

Save and recall parameter configurations as JSON files.
Preset files are automatically stored in the `Assets/UniTexEditor_Presets/` folder.

Saving a Preset

1. Enter a name in the **Name** field
2. Select the **Type**:
   - **Parameters only**: Saves all numeric values, toggles, and curves — no texture references
   - **Include Textures**: Also saves references to the mask and blend textures (by GUID, so renames and moves within Unity are handled safely)
3. Click the **Save** button
   - If a preset with the same name already exists, a confirmation dialog will ask whether to overwrite it

> **Note**: The source texture is never included in a preset. When loading, whichever source texture is currently set remains unchanged.

Loading a Preset

1. Select a preset from the dropdown — the info line shows its type, save date, and how many sections are active
2. Click **Load**
   - For texture-inclusive presets, if any referenced texture cannot be found, a warning is shown but all other parameters are still applied

Deleting a Preset

Select a preset from the dropdown and click **Delete**. A confirmation dialog will appear before the file is removed.

> ↺ **Button**: Manually refreshes the preset list — useful if you added files to the preset folder outside of Unity.

Color Variation Generator (CVG)

Automatically generates color variants from the current preview result.

- **Hue Steps**: Number of divisions around the hue wheel (1 to 36)
- **Saturation Steps**: Number of saturation levels (1 to 10)
- **Generate Grayscale**: Also generate a saturation-zero variant
- **Output**: Destination folder (defaults to the same folder as the source image)

Generated files are saved in a folder named `[source_name]_variations_[date]/`.

Language

Click the **JA** / **EN** button in the upper-right corner of the editor window to switch between Japanese and English.

Output Settings

Choose how to save the result at the bottom of the window.

- **Overwrite Source**: Overwrite the original file directly (**Warning**: this cannot be undone)
- **Select...**: Choose a new file path to save as a separate file

The output is always converted to the sRGB color space, ensuring correct display in standard image viewers.

Tips

- **Auto-update**: Enable "Auto Update" in the preview area to refresh the preview automatically on every parameter change.
- **Reset Section**: Each section has a "Reset" button to restore its parameters to defaults.
- **Reset All**: The button at the bottom of the window resets every parameter to its initial state.

Requirements

- Unity 2022.3.22f1 or later
- Compute Shader-compatible GPU (DirectX 11, Metal, etc.)